

Robert I. Booth

robert.booth@cs.ox.ac.uk

British and French nationality

My research focusses on the mathematical foundations of quantum computing. In no particular order, I have worked on: diagrammatic reasoning, measurement-based quantum computation, measures of non-classicality in quantum theory, and quantum computational with continuous variables. Some topics that particularly excite me at the moment are fault-tolerant quantum compilation and the design and semantics of quantum programming languages.

Research experience

06/2025–05/2027 <i>ongoing</i>	Research Associate in Quantum Software Department of Computer Science, University of Oxford Working with Aleks Kissinger • <i>Quantum Compilation</i> • <i>Quantum Fault-tolerance</i> • <i>Quantum programming languages and semantics</i> • <i>Diagrammatic reasoning</i>
04/2022–05/2025	Research Associate School of Informatics, University of Edinburgh School of Mathematics, University of Bristol Working with Chris Heunen and Noah Linden • <i>Optimisation and distribution of quantum circuits</i> • <i>Resources for distributed quantum computation.</i> • <i>Diagrammatic reasoning and applied category theory</i>
11/2018–02/2022	Doctoral Student in Quantum Computation Laboratoire d'Informatique de Paris 6, Sorbonne Université, Paris Laboratoire Lorrain de Recherche en Informatique et ses Applications, Nancy Title: <i>Measurement-based quantum computation beyond qubits</i> Supervised by Damian Markham and Simon Perdrix • <i>Measurement-based quantum computation with qudits and continuous variables</i> • <i>Quantum circuit extraction algorithms and the qudit ZX-calculus</i> • <i>Contextuality in continuous-variable systems and its connection with other notions of non-classicality</i>
Mar.–Jul. 2018	Research Intern in Quantum Information Theory Laboratoire d'Informatique de Paris 6, Université Pierre et Marie Curie, Paris Supervised by Damian Markham
Apr.–Jul. 2017	Research Intern in Experimental Quantum Optics Università degli Studi Roma Tre, Rome Supervised by Marco Barbieri
Feb.–May 2016	Research Intern in Experimental Physical Chemistry Laser Chemistry, Spectroscopy and Dynamics Laboratory University of Bristol, United Kingdom Supervised by Prof. Michael N. R. Ashfold FRS

Teaching experience

Spring 2024	Introduction to Quantum Programming and Semantics <i>Guest Lectures and Tutorials</i> University of Edinburgh
Spring 2023	Categories and Quantum Informatics <i>Lectures, Computer Labs, and preparation of course material</i> University of Edinburgh <i>My work on this course earned a University of Edinburgh Staff Award</i>

Education

2016–2018	Master's Degree in Fundamental Physics Université Pierre et Marie Curie, Paris
2016	Double Honours Bachelor's Degree in Physics and Chemistry Université Paris-Sud, Orsay, France
2012	Scientific Baccalaureate , France

Students

Nathan Hall (MSc 2026), Axel Daboust (MSc 2026), Sascha Haupt (MSc 2024–2026), Boren Gu (MSc 2025, Freie Universität Berlin), Clément Poirson (MSc 2024, Inria / Alice & Bob)

Publications

Journal articles

- Francesco Arzani, Robert I. Booth, and Ulysse Chabaud (Nov. 6, 2025). “Effective descriptions of bosonic systems can be considered complete”. In: *Nature Communications* 16.1. Publisher: Nature Publishing Group, p. 9744. DOI: [10.1038/s41467-025-64872-3](https://doi.org/10.1038/s41467-025-64872-3)
- Robert I. Booth and Damian Markham (Oct. 19, 2023). “Flow conditions for continuous variable measurement-based quantum computing”. In: *Quantum* 7. Publisher: Verein zur Förderung des Open Access Publizierens in den Quantenwissenschaften, p. 1146. DOI: [10.22331/q-2023-10-19-1146](https://doi.org/10.22331/q-2023-10-19-1146)
- Robert I. Booth, Aleks Kissinger, Damian Markham, Clément Meignant, and Simon Perdrix (Feb. 2023). “Outcome determinism in measurement-based quantum computation with qudits”. In: *Journal of Physics A: Mathematical and Theoretical* 56.11. Publisher: IOP Publishing, p. 115303. DOI: [10.1088/1751-8121/acbace](https://doi.org/10.1088/1751-8121/acbace)
- Robert I. Booth, Ulysse Chabaud, and Pierre-Emmanuel Emeriau (Nov. 29, 2022). “Contextuality and Wigner negativity are equivalent for continuous-variable quantum measurements”. In: *Physical Review Letters* 129.23, p. 230401. DOI: [10.1103/PhysRevLett.129.230401](https://doi.org/10.1103/PhysRevLett.129.230401). arXiv: [2111.13218](https://arxiv.org/abs/2111.13218)[[math-ph](#), [physics:quant-ph](#)]
- Vasco Cavina, Luca Mancino, Antonella De Pasquale, Ilaria Gianani, Marco Sbroscia, Robert I. Booth, Emanuele Roccia, Roberto Raimondi, Vittorio Giovannetti, and Marco Barbieri (Nov. 9, 2018). “Bridging thermodynamics and metrology in nonequilibrium quantum thermometry”. In: *Physical Review A* 98.5. DOI: [10.1103/PhysRevA.98.050101](https://doi.org/10.1103/PhysRevA.98.050101)
- Luca Mancino, Vasco Cavina, Antonella De Pasquale, Marco Sbroscia, Robert I. Booth, Emanuele Roccia, Ilaria Gianani, Vittorio Giovannetti, and Marco Barbieri (Oct. 17, 2018). “Geometrical Bounds on Irreversibility in Open Quantum Systems”. In: *Physical Review Letters* 121.16. DOI: [10.1103/PhysRevLett.121.160602](https://doi.org/10.1103/PhysRevLett.121.160602)

Conference proceedings

- Robert I. Booth and Cole Comfort (2026). “Denotational Semantics for Stabiliser Quantum Programs”. In: *11th International Conference on Formal Structures for Computation and Deduction (FSCD 2026)*. Ed. by Frank Pfenning. Vol. 378. Leibniz International Proceedings in Informatics (LIPIcs). ISSN: 1868-8969. Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 8:1–8:24. DOI: [10.4230/LIPIcs.FSCD.2026.8](https://doi.org/10.4230/LIPIcs.FSCD.2026.8)
- Robert I. Booth, Titouan Carrette, and Cole Comfort (2026). “Graphical Symplectic Algebra”. In: *11th International Conference on Formal Structures for Computation and Deduction (FSCD 2026)*. Ed. by Frank Pfenning. Vol. 378. Leibniz International Proceedings in Informatics (LIPIcs). ISSN: 1868-8969. Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 7:1–7:22. DOI: [10.4230/LIPIcs.FSCD.2026.7](https://doi.org/10.4230/LIPIcs.FSCD.2026.7)

- Boldizsár Poór, Robert I. Booth, Titouan Carette, John Van De Wetering, and Lia Yeh (Aug. 29, 2023). “The Qupit Stabiliser ZX-travaganza: Simplified Axioms, Normal Forms and Graph-Theoretic Simplification”. In: *Electronic Proceedings in Theoretical Computer Science*. Twentieth International Conference on Quantum Physics and Logic. Vol. 384. Paris, France, pp. 220–264. DOI: [10.4204/EPTCS.384.13](https://doi.org/10.4204/EPTCS.384.13)
- Robert I. Booth and Titouan Carette (2022). “Complete ZX-Calculi for the Stabiliser Fragment in Odd Prime Dimensions”. In: *47th International Symposium on Mathematical Foundations of Computer Science (MFCS 2022)*. Ed. by Stefan Szeider, Robert Ganian, and Alexandra Silva. Vol. 241. Leibniz International Proceedings in Informatics (LIPIcs). ISSN: 1868-8969. Dagstuhl, Germany: Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 24:1–24:15. DOI: [10.4230/LIPIcs.MFCS.2022.24](https://doi.org/10.4230/LIPIcs.MFCS.2022.24)

Preprints

- Alex Rice, Dominik Leichtle, Kim Worrall, and Robert I. Booth (July 10, 2026). *Quantum Orchestras: a Concrete Semantics for Recursive Hybrid Programs*. DOI: [10.48550/arXiv.2607.09605](https://doi.org/10.48550/arXiv.2607.09605). arXiv: [2607.09605\[cs.PL\]](https://arxiv.org/abs/2607.09605)
- Robert I. Booth, Dominik Leichtle, Alex Rice, and Kim Worrall (June 26, 2026). *Composing Quantum Instruments*. DOI: [10.48550/arXiv.2606.28291](https://doi.org/10.48550/arXiv.2606.28291). arXiv: [2606.28291\[quant-ph\]](https://arxiv.org/abs/2606.28291)
- Robert I. Booth, Titouan Carette, and Cole Comfort (Mar. 15, 2024). *Complete equational theories for classical and quantum Gaussian relations*. arXiv: [2403.10479\[quant-ph\]](https://arxiv.org/abs/2403.10479)
- Clément Poirson, Joschka Roffe, and Robert I. Booth (May 5, 2025). *Engineering CSS surgery: compiling any CNOT in any code*. DOI: [10.48550/arXiv.2505.01370](https://doi.org/10.48550/arXiv.2505.01370). arXiv: [2505.01370\[quant-ph\]](https://arxiv.org/abs/2505.01370)

Dissemination

Conferences

- “Can effective descriptions of bosonic systems be considered complete?”, 29th Annual Quantum Information Processing Conference, Riga, Latvia, 2026 (Francesco Arzani)
- “CSS surgery: compiling any CNOT in any code”, 22nd International Conference on Quantum Physics and Logic, Varna, Bulgaria, 2025
- “A graphical language for Gaussian quantum processes”, 21st International Conference on Quantum Physics and Logic, Buenos Aires, Argentina, 2024
- “Contextuality and Wigner negativity are equivalent for continuous-variable measurements”, 19th International Conference on Quantum Physics and Logic, Oxford, 2022 (Pierre-Emmanuel Emeriau)
- “A graphical language for Gaussian quantum processes”, 21st International Conference on Quantum Physics and Logic, Buenos Aires, Argentina, 2024
- “Complete equational theories for classical and quantum Gaussian relations”, 7th International Conference on Applied Category Theory, Oxford, 2024 (Cole Comfort)
- “Graphical Symplectic Algebra”, 7th International Conference on Applied Category Theory, Oxford, 2024 (Cole Comfort)
- “The Qupit Stabiliser ZX-travaganza: Simplified Axioms, Normal Forms, and Graph-theoretic Simplification”, 20th International Conference on Quantum Physics and Logic, Paris, 2023 (Boldizsár Poór)
- “Complete ZX-calculi for the stabiliser fragment in odd prime dimensions”, 47th International Symposium on Mathematical Foundations of Computer Science, Vienna, 2022
- “Complete ZX-calculi for the stabiliser fragment in odd prime dimensions”, 19th International Conference on Quantum Physics and Logic, Oxford, 2022 Video: [YouTube](https://www.youtube.com/watch?v=...)
- “Contextuality and Wigner negativity are equivalent for continuous-variable measurements”, 19th International Conference on Quantum Physics and Logic, Oxford, 2022 (Pierre-Emmanuel Emeriau) Video: [YouTube](https://www.youtube.com/watch?v=...)
- “Extracting reversible quantum circuits from measurement-based quantum computations with qudits”, 19th International Conference on Quantum Physics and Logic, Oxford, 2022 Video: [YouTube](https://www.youtube.com/watch?v=...)

- “Contextuality and Wigner negativity are equivalent for continuous-variable measurements”, 20th European Conference on Foundations of Physics, Ecole Normale Supérieure Paris, October 30th, 2021
- “F-flow: determinism in measurement-based quantum computing with qudits”, 19th International Conference on Quantum Physics and Logic, University of Gdansk, June 9th, 2021, Video: [YouTube](#)
- “F-flow: determinism in measurement-based quantum computing with qudits”, Sixth International Conference for Young Quantum Information Scientists, Michigan State University, April 15th, 2021
- “Flow conditions for continuous variable measurement-based quantum computation”, 18th International Conference on Quantum Physics and Logic, Paris, June 15th, 2020, Video: [YouTube](#)

Workshops

- “Denotational semantics for stabiliser quantum programs”, 6th International Workshop on Programming Languages for Quantum Computing, Rennes, France, 2026 (Cole Comfort), Video: [YouTube](#)
- “Denotational semantics for stabiliser quantum programs”, 6th Scottish Programming Language Seminar, Glasgow, United Kingdom, 3 December 2025
- “A symplectic vision for the ZX-calculus”, CATNIP workshop, University of Aberdeen, May 3rd, 2024
- “Distributing quantum circuits with the ZX-calculus”, National Center for Quantum Computing Launch Event, University of Edinburgh, December 13th, 2022
- “Outcome determinism in measurement-based quantum computing with qudits”, Workshop on recent advances on quantum computing, Collège de France, June 17th, 2021
- “Flow conditions for continuous-variable measurement-based quantum computation”, Workshop on Quantum Networks and Information 2020, National Institute of Informatics, Tokyo
- “Convergence of continuous-variable measurement-based quantum computations”, Edinburgh-Paris Joint Quantum Workshop, Edinburgh, July, 2019

Seminars

- “The algebra for quantum fault-tolerance”, Algebraic and Combinatorial Perspectives in the Mathematical Sciences (online), 22 May 2026
- “A convenient setting for fault-tolerant compilation”, Quantum Lunch seminar, University of Oxford, 30 April 2025
- “Finding a convenient setting for fault-tolerant quantum compilation”, University College London, 29 April 2025
- “CSS surgery: compiling any CNOT in any CSS code”, PhiQus team seminar, École Polytechnique, Saclay, France, 28 November 2024
- “The nullifier theory of Gaussian quantum states”, Qatalyze seminar, Paris, France, 27 November 2024
- “Complete Equational Theories for Classical and Quantum Gaussian Relations”, ZX seminar, 2024, online, Video: [YouTube](#)
- “Graphical reasoning for Gaussian quantum mechanics”, Loria, Nancy, 9 April 2024
- “Graphical Symplectic Algebra”, ZX seminar, 2024, online, Video: [YouTube](#)
- “ZX-calculus: (my) past, present... future?”, LIP6 QI team seminar, Paris, 2023
- “Complete ZX-calculi for the stabiliser fragment in odd prime dimensions”, ZX seminar, 2022, online, Video: [YouTube](#)
- “Contextuality and Wigner negativity are equivalent for continuous-variable measurements”, Online Quantum and Linear Optical Computation seminar, 2022
- “Outcome determinism in measurement-based quantum computing with qudits”, ZX seminar, 2021, online
- “F-flow: determinism in measurement-based quantum computing with qudits”, ZX seminar, 2020, online

Academic service

- Organiser of the University of Edinburgh's Quantum Software Lab Seminar, 2023-2025
- Member of the Program Committee for *QPL 2026*, *QPL 2025*, *ACT 2025*, *PLanQC 2022*
- Reviewer for the US Department of Energy grant call "Express: 2023 Exploratory Research for Extreme-Scale Science"
- Reviewer for the journals *Quantum*, *Compositionality*, *Logical Methods in Computer Science*
- Reviewer for the conferences *TQC 2026*, *LICS 2026*, *LICS 2025*, *ACT 2024*, *QPL 2023*, *QPL 2022*, *QPL 2021*
- Held the "Quantum Spies" stand at Edinburgh Science Festival 2022, and alongside the Doctor Who Worlds of Wonder exhibition, both at the National Museum of Scotland

Other

Languages | **English and French, bilingual**

Computing | Python, Mathematica, UNIX, $\text{\LaTeX}$